FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. BD

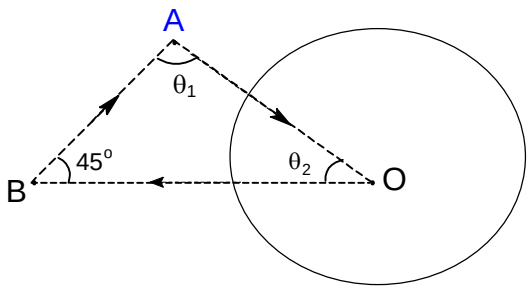
Sol. $\frac{\left(\sqrt{3}+1\right)R}{\sin\theta_1}=\frac{2R}{\sin 45^\circ}$

$\theta_1=105^\circ \ or \ 75^0$

$so \ \theta_2=30^0 \ or \ 60^0$

$\oint\limits_{OBAO}\vec{B}\cdot\vec{d\ell}=\mu_0 I_{in}$

$\oint\limits_{BA}\vec{B}\cdot\vec{d\ell}=\frac{-\mu_0 I}{6} \quad or \quad \frac{-\mu_0 I}{12}$



2. BC

Sol. For mirror, from mirror formula

$\frac{1}{V}+\frac{1}{-10}=\frac{1}{-20}\Rightarrow V=+20$

$m=\frac{-V}{u}=2$

For spherical mirror

$v_1=-m^2 \ v_0=-4\times 1=-4 \ cm/s$

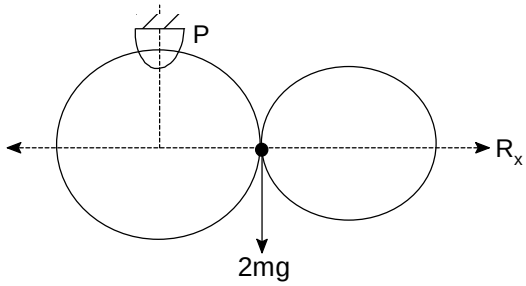
Now using concept of apparent depth

$v_{IB}=1cm/s$ along downward

$v_{IF} = 5cm/s$ along downward

3. BD

Sol. For Bead



$2mgR - R_xR = 8mR^2\alpha$ (1)

$R_x = ma_x$ (2)

$a_x = \alpha R$ (3)

From (1), (2) & (3)

$\alpha = \dfrac{2g}{9R}$

$mg = ma_y \Rightarrow a_y = g$

$acceleration\ of\ bead = \dfrac{\sqrt{85}g}{9}$

4. ABD

Sol. Condition of no. slipping $\omega R - 6\omega_0 R = 4\omega_0 R \qquad \omega = 10\ \omega_0$

Acceleration of Point A

$\vec{a}_A = \vec{a}_{A,C} + \vec{a}_C \qquad \uparrow 12\omega_0^2 R$

$a_A = 88\omega_0^2 R \downarrow \qquad \overset{\bullet A}{\downarrow} (10\omega_0)^2 R$

Radius of curvature

$\rho = \dfrac{v_A^2}{a_n} = \dfrac{(16\omega_0 R)^2}{88\omega_0^2 R} = \dfrac{32R}{11}$

Acceleration of point B with respect to surface $\left(\vec{a}_B - \vec{a}_{Surface}\right)$,

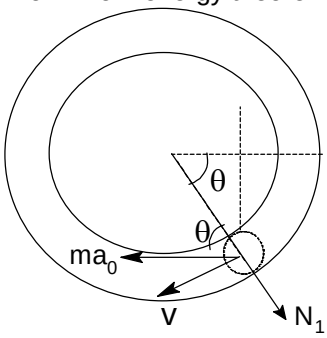
$\vec{a}_B = \vec{a}_{B,C} + \vec{a}_C$

$\vec{a}_S = \omega_0^2 4R = 4\omega_0^2 R$

$\vec{a}_{PS} = (112 - 4)\omega_0^2 R = 108\omega_0^2 R$

5. AD

Sol. From Work energy theorem



$$ma_0R(1-\cos\theta)=\frac{1}{2}mv^2$$

$$v^2=2a_0R(1-\cos\theta) \quad ---(1)$$

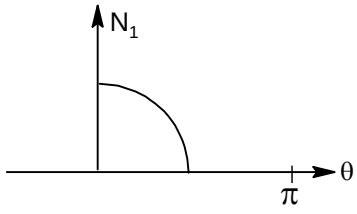
$$ma_0\cos\theta-N_1=\frac{mv^2}{R} \quad ---(2)$$

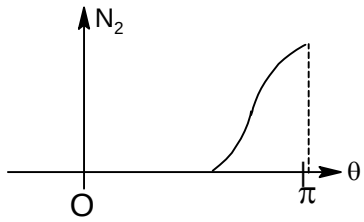
$$N_1=(3\cos\theta-2)ma_0$$

$$when\ \cos\theta=\frac{2}{3} \qquad N_1=0$$

$$for\ \theta>\cos^{-1}\left(\frac{2}{3}\right) \qquad N_1=0$$

$$N_2=ma_0(2-3\cos\theta)$$





6. ABCD

Sol. Let temperature of A, B and C are $T_1, T_2\ and\ T_3$ respectively. Then

Equating power for C

$$T_2=2T_3 \quad ---(1)$$

$$T_1^4=(256-64)T_3^4$$

$$T_1=2\sqrt{2}\ 3^{\frac{1}{4}}T_3 \quad ---(2)$$

$$T_1^4+64T_3^4=16T_2^4 \quad ---(3)$$

$$\frac{i^2\rho\ell}{\pi r^2}+\sigma 2\pi 8r\ell T_2^4=\sigma 2\pi r\ell T_1^4 \ .---(4)$$

7. ABCD

Sol. Normal force will balance pressure force exerted along horizontal direction. Weight of object is equal to pressure force exerted along vertical direction.

$N = \frac{\rho g r^2 \ell}{\ell}$

$w = \left(\sqrt{3} + 1 + \frac{\pi}{4}\right) r^2 \rho \ell g$

8. D

Sol. use induced emf for a moving conductor $E = \left(\vec{V} \times \vec{B}\right).\overrightarrow{\Delta \ell}$

9. C

Sol. use induced emf for a moving conductor $E = \left(\vec{V} \times \vec{B}\right).\overrightarrow{\Delta \ell}$

10. D

Sol. use induced emf for a moving conductor $E = \left(\vec{V} \times \vec{B}\right).\overrightarrow{\Delta \ell}$

11. D

Sol. use the concept net field inside conducting plate is zero and for isolated plate the charge is conserved. When plate A and C are connected by conducting wire, the potential difference between A and C is zero.

12. B

Sol. use the concept net field inside conducting plate is zero and for isolated plate the charge is conserved. When plate A and C are earthed, the potential difference between A and C is zero.

13. A

Sol. use the concept net field inside conducting plate is zero and for isolated plate the charge is conserved. The sum of charge on plate A and C is constant.

SECTION – B

14. 3

Sol. Disc will move translationally if line of action of force passes through c.m of semi circular disc.

$y_{cm} = \frac{\int y\, dm}{\int dm} = \frac{3\pi R}{16}$

So value of $K = 3$

15. 8 m/s

Sol. Acceleration of particle with respect to box is along perpendicular to inclined plane

$v_{\min} = \sqrt{g \cos 37^\circ \left(AB + CD + AC\right)}$

$= \sqrt{8\left(2 + 3 + 3\right)} = 8\, m/s$

16. 5m

Sol. Using Bernoulli's equation and continuity equation

$\ell' = \frac{A}{a}\sqrt{\frac{\rho A}{4\mu}}\ell$

Where H is length of Can $\ell' = 5\, m$

17. 5

Sol. From Gauss law for spherical space charge density

$$\frac{dE}{dr}+\frac{2E}{r}=\frac{\rho}{\in_0}$$

$$\rho = 5K \in_0 r^2$$

$$\left.\frac{d\rho}{dr}\right|_{r=\frac{1}{2}m} = 5K \in_0$$

So, value of α is 5

18. 2

Sol. Force on dielectric due to induced charge is $F=\frac{1}{2}$

From Force and Torque equation on cylinder and dielectric

The acceleration of dielectric will be $6\,m/s^2$

$$I=\frac{\in_0 a(K-1)E}{d}v$$

$$\frac{dI}{dt}=\frac{\in_0 a(K-1)E}{d}\frac{dv}{dt}=2$$

Chemistry

PART – II

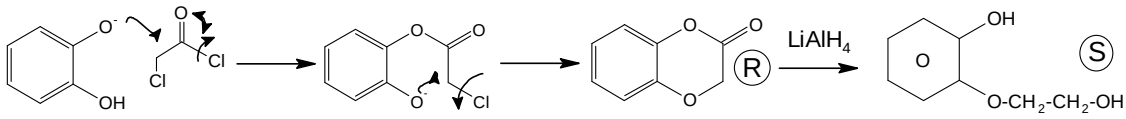
SECTION – A

19. ABC

Sol. In option – D, Acid / base reaction will take place

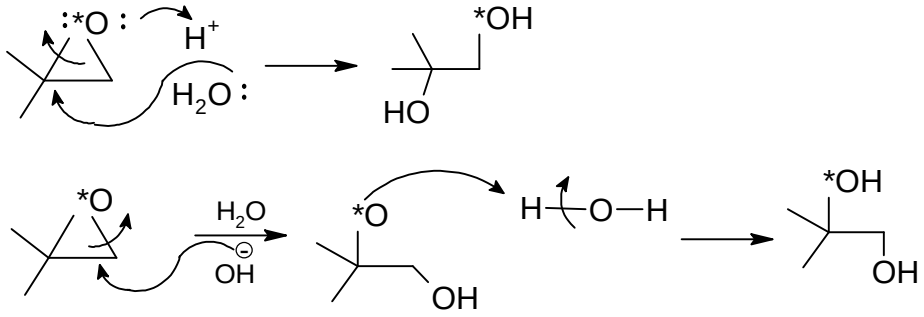
20. BCD

Sol.



21. ABC

Sol.



22. ABCD

Sol. $k_{eq} = 4, \frac{k_1}{k_2}$

$so,\ k_2 = 10^{-2}$

$t\,o.\mathrm{s} = 28\ \mathrm{sec}$

23. AC

Sol. $B.O\ of\ NO^+ = 3{,}16,\ 2\pi\ bond$

$BO\ of\ NO = 2.5$

$Both\ NO^+,\ BF^-\ is\ \mathrm{diam}\,agnetic$

B – F bonds are longer in $BF_4^-\ than\ in\ BF_3$ due to absence at $P\pi - d\pi\ B\cdot B\ in\ \left[BF_4\right]^-$

24. ACD

Sol. Due to presence of SCN^- it will show linkage isomerism. It will be optically inactive as plane of symmetric is present. IT will also show geometrical isomerism and polymerization.

25. AC

Sol.

$n\lambda = 2\pi r \qquad so,\ \lambda_3 = \frac{2\pi \times 3}{\quad} \times x \Rightarrow 6\pi x$

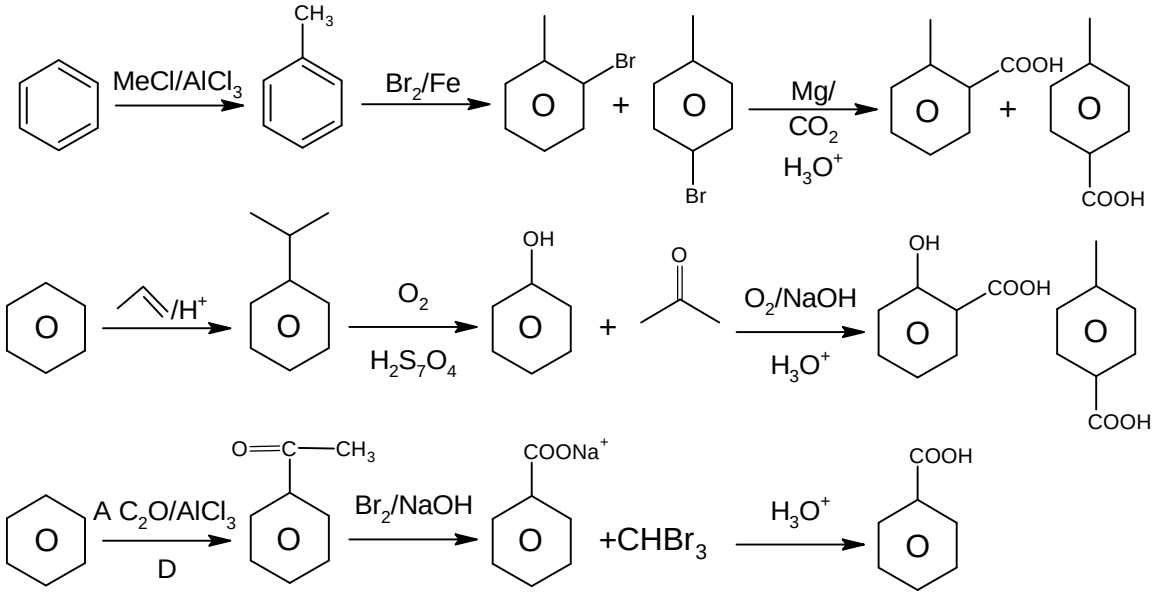
$\lambda_3\left(11^{2+}\right) = 2\pi x$

26. A
Sol. Conceptual

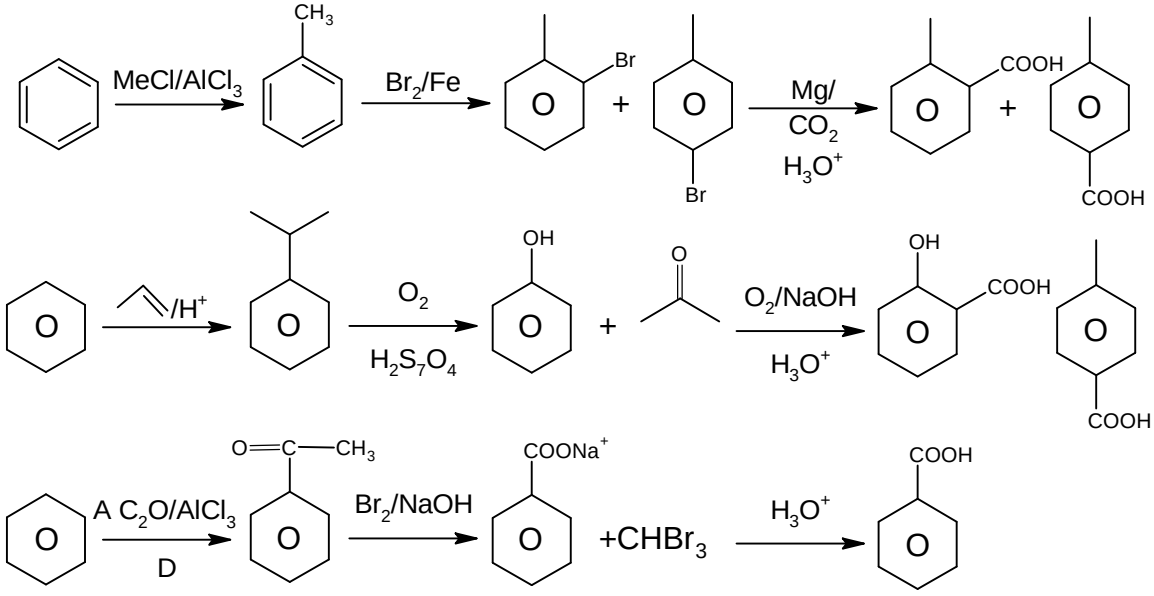
27. B
Sol. Conceptual

28. C
Sol. Conceptual

29. B
Sol.

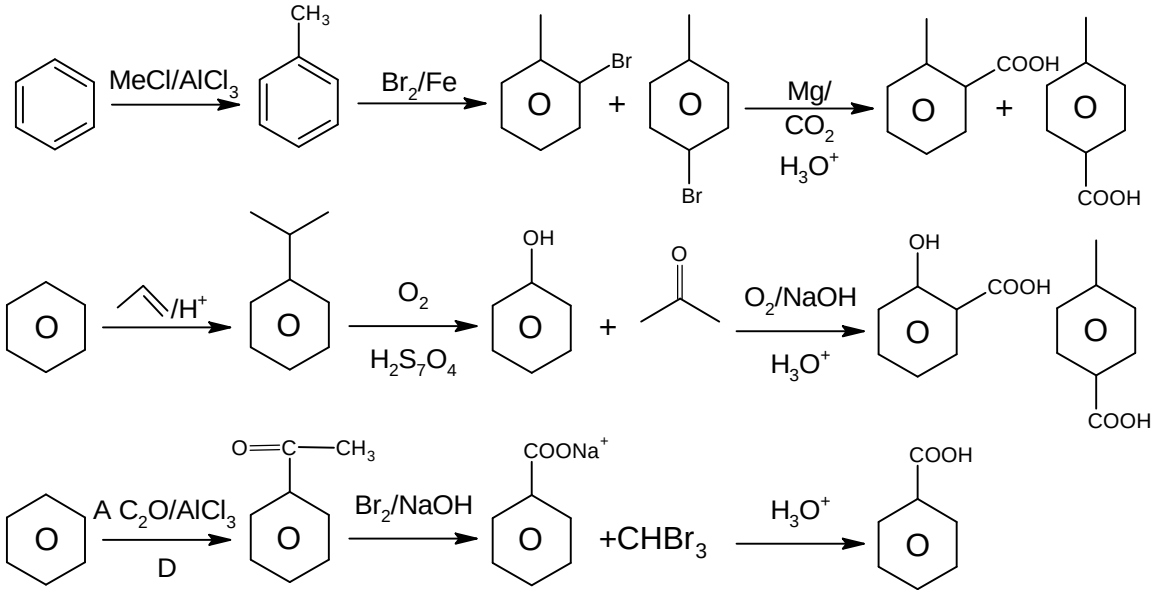


30. A
Sol.



31. C

Sol.



SECTION – B

32. 5

Sol. $kp = \left(P_{H_{20}}\right)^2 = P_{H_{20}} = 15\times10^{-3}$

$$R.H = \frac{P_{H_{20}}}{S.U.P} = \frac{15\times10^{-3}}{\frac{22.8}{760}} = 0.5$$

$$\%\,RH = 50\%,\ \ so,\ \frac{x}{10} = 5$$

33. 3

Sol. $f = e^{\frac{-Ea}{RT}} \Rightarrow \frac{10^{-3}}{100} = 10^{-5}$

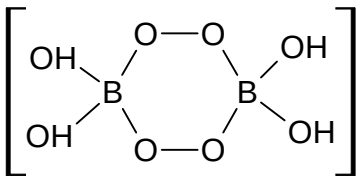
$x = Ae^{\frac{-Ea}{RT}}$

$3\times10^{-3} = A\times10^{-5} \Rightarrow A = 300\,Min^{-1}$

Maximum value of rate constant is equal to A is

34. 7

Sol. $B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 6H_2$



35. 7

Sol. Number of Faraday's $= \frac{1.25 \times 212 \times 60}{96000} \Rightarrow 0.165$

$\Rightarrow 0.16$ *mol of* HPO_4^- will be converted in to $H_2PO_4^-$ because 0.16 *mol* H^+ will be produced at anode

So, $pH = Pka + \log \frac{(1-0.16)}{(1+0.16)} \Rightarrow 7.1 - 0.1 = 7$

36. 4

Sol. $An_{(g)} \rightleftharpoons nA(g)$

$100 \qquad 0$

$48 \qquad 52n$

$M\ mix = \frac{48}{48+52n} \times 32n + \frac{n \times 52}{48+2n} \times 32$

$= \frac{32n \times 100}{48+52n}$

now, $\frac{1}{1.25} = \sqrt{\frac{32}{\frac{32n \times 100}{48+52n}}}$

on solving we get $n = 4$

Mathematics

PART – III

SECTION – A

37. AB

Sol. 1 $\perp$ distance of P(x, y, z) from $z - ax = \sqrt{x^2 + y^2}$

38. ABC

Sol. $r_1 = \sqrt{(\alpha)^2 - \gamma},\ r_2 = \sqrt{(\beta)^2 - \delta}$

External touch $c_1c_2 = r_1 + r_2$

c_1 & c_2 is orthogonal if $|\alpha - \beta|^2 = |\alpha|^2 - \gamma + |\beta|^2 - \delta$

39. AD

Sol. $\frac{\pi}{2} > 1$ & $\frac{\pi}{2} < 2$

$x^{\frac{\pi}{2}} < x^1 \ (0 < x < 1)$ & $x^{\frac{\pi}{2}} > x^2 \ (0 < x < 1)$

$\int_0^1 \frac{1}{1+x^{\frac{\pi}{2}}} > \int_0^1 \frac{1}{1+x}$ & $\int_0^1 \frac{1}{1+x^{\frac{\pi}{2}}} < \int_0^1 \frac{1}{1+x^2}$

$I > \ell n_2$ & $I < \frac{\pi}{4}$

40. BD

Sol. Let $(\lambda, 1)$ is a point on the given curve

$\frac{dy}{dx} = -\frac{a}{\lambda} \ (at\ \lambda, 1)$

$y - 1 = -\frac{a}{\lambda}(x - \lambda)$

$Now,\ x = 0$

$y = 1 + a$

$y = 0$

$x = \frac{\lambda}{a} + \lambda = \frac{\lambda(1+a)}{a}$

$\frac{dA}{da} = \frac{1}{2}\left(\frac{a \cdot 2 \cdot (1+a) - (1+a)^2}{a^2}\right) \cdot \lambda = 0$

$a = 1, -1$

41. ACD

Sol. $px - y = \sin^{-1} p$

$$\left(x-\frac{1}{\sqrt{1-p^2}}\right)\frac{dp}{dx}=0 \Rightarrow p=c \ \& \ x=\frac{1}{\sqrt{1-p^2}}$$

42. BD

Sol. $h'(x)=f'(x)-2f(x)f'(x)+3f^2(x)+f'(x)$

$$f'(x)\left[1-\underbrace{2f(x)}_{+ve}+3\left(f(x)\right)^2\right]$$

43. AD

Sol. $$p=\frac{3nc_n}{2nc_n}=\lim_{n\to\infty}\left[\left(1+\frac{n}{n+1}\right)\left(1+\frac{n}{n+2}\right)-----\left(1+\frac{n}{n+n}\right)\right]^{\frac{1}{n}}$$

$$\ell n\, p=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ell n\left(1+\frac{1}{1+r/n}\right)$$

$$\ell n\, p=\int_0^1 \ell n\left(\frac{x+2}{x+1}\right)$$

$$p=\frac{27}{16}$$

44. B

Sol. I.

$$I'(\alpha)=\int_0^1\frac{x^\alpha \ell n\, x}{\ell n\, x}$$

$$I'(\alpha)=\ell n\left|\alpha+1\right|$$

II.

$$I'(\alpha)=\int_0^1\frac{x\,dx}{x\sqrt{1-x^2}\left(1+a^2x^2\right)}$$

$$I(\alpha)=\frac{\pi}{2}\ell n\left|\alpha+\sqrt{1+\alpha^2}\right|$$

III.

$$I'(\alpha)=\int_0^1\frac{-2\alpha x^2}{\left(1-a^2x^2\right)\cdot x^2\sqrt{1-x^2}}dx$$

$$I(\alpha)=\pi\left|\sqrt{1-\alpha^2}-1\right|$$

IV.

$$I(\alpha)=\int_0^\infty\frac{ta^2\alpha x}{x\left(1+x^2\right)}dx$$

$$I'(\alpha)=\int_0^\infty\frac{1}{1+\alpha^2x^2}\cdot\frac{x}{x\left(1+x^2\right)}dx$$

$$I(\alpha)=\frac{\pi}{2}\ell n\left(\alpha+1\right)$$

45. C

Sol. I.

$$I'(\alpha)=\int_0^1 \frac{x^\alpha \ell n\, x}{\ell n\, x}$$

$$I'(\alpha)=\ell n|\alpha+1|$$

II.

$$I'(\alpha)=\int_0^1 \frac{x\,dx}{x\sqrt{1-x^2}\left(1+a^2x^2\right)}$$

$$I(\alpha)=\frac{\pi}{2}\ell n\left|\alpha+\sqrt{1+\alpha^2}\right|$$

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$$I'(\alpha)=\int_0^1 \frac{-2\alpha x^2}{\left(1-a^2x^2\right)\cdot x^2\sqrt{1-x^2}}dx$$

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IV.

$$I(\alpha)=\int_0^\infty \frac{ta^2\alpha x}{x\left(1+x^2\right)}dx$$

$$I'(\alpha)=\int_0^\infty \frac{1}{1+\alpha^2x^2}\cdot\frac{x}{x\left(1+x^2\right)}dx$$

$$I(\alpha)=\frac{\pi}{2}\ell n(\alpha+1)$$

46. C

Sol. I.

$$I'(\alpha)=\int_0^1 \frac{x^\alpha \ell n\, x}{\ell n\, x}$$

$$I'(\alpha)=\ell n|\alpha+1|$$

II.

$$I'(\alpha)=\int_0^1 \frac{x\,dx}{x\sqrt{1-x^2}\left(1+a^2x^2\right)}$$

$$I(\alpha)=\frac{\pi}{2}\ell n\left|\alpha+\sqrt{1+\alpha^2}\right|$$

III.

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IV. $I(\alpha)=\int_0^{\infty}\frac{ta^2\alpha x}{x\left(1+x^2\right)}dx$

$I'(\alpha)=\int_0^{\infty}\frac{1}{1+\alpha^2x^2}\cdot\frac{x}{x\left(1+x^2\right)}dx$

$I(\alpha)=\frac{\pi}{2}\ell n(\alpha+1)$

47. C

Sol. $(x-1)^2+(y-2)^2=\alpha(x+y-9)^2$

$e^2\left(\frac{x+y-9}{\sqrt{2}}\right)^2$

$\alpha=e^2/2$

48. B

Sol. $(x-1)^2+(y-2)^2=\alpha(x+y-9)^2$

$e^2\left(\frac{x+y-9}{\sqrt{2}}\right)^2$

$\alpha=e^2/2$

49. D

Sol. $(x-1)^2+(y-2)^2=\alpha(x+y-9)^2$

$e^2\left(\frac{x+y-9}{\sqrt{2}}\right)^2$

$\alpha=e^2/2$

SECTION – B

50. 6

Sol. $\left(y^2-3\right)^2+(x-4)^2=1$

$x=4+\cos\theta,\ y^2=3+\sin\theta$

$M=36,\ m=1$

51. 8

Sol. If e & e_1 are eccentricity of hyperbola & in conjugate

$\frac{1}{e^2}+\frac{1}{e_1^2}=1 \quad \Rightarrow f(e)=\frac{e}{\sqrt{e^2-1}} \quad \Rightarrow fof(e)=e$

$$\underbrace{fofo---f(e)}_{n\ time}=\begin{cases} e & if\ \ n\ \ is\ \ even \\ \frac{e}{\sqrt{e^2-1}} & if\ \ n\ \ is\ \ odd \end{cases}$$

$$=\begin{cases} 4 & if\ \ n\ \ is\ \ even \\ 2\sqrt{2} & if\ \ n\ \ is\ \ odd \end{cases}$$

52. 1

Sol. $Let\ y=x^2-2=g(x)$

$\Rightarrow x=\sqrt{y+2}$

$p(x)=0\Rightarrow p\left(\sqrt{y+2}\right)=0$

$\left(\sqrt{y+2}\right)^5+\left(\sqrt{y+2}\right)^2+1=0$

$y^5+20y^4+40y^3+79y^2+74y+23=0$

53. 3

Sol. $Let\ h=4+4\cos\phi,\ k=3+3\sin\phi$

Reflection about line $x-y-z=0$ & taken 10cm of (h, k)

We get,

$16x^2+9y^2-160x-36y+292=0$

$k_1+k_2=132=2^2\cdot3\cdot11$

54. 9

Sol. $\sin x+\cos y=-2$

$\sin x=-1\ \ \&\ \cos y=-1$

$x=-\frac{\pi}{2}\ \&\ y=\pi\Rightarrow\left(x^2+y^2\right)\min$

$=\frac{\pi^2}{4}+\pi^2=\frac{5}{4}\pi^2$